

AI-Powered Churn Prevention

A Data-Driven Business Proposal for Company A

Dataset Company A — Telecom (100,000 customers)

Scope

EDA · Machine Learning · Business Proposal

Models

Logistic Regression · Random Forest · Gradient Boosting

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100K customers, 100 features explored

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The Telecom Churn Problem

Why Churn Matters

Customer churn is the #1 profitability threat for telecom operators worldwide. **Acquiring a new customer costs 5–7× more than retaining an existing one** (Bain & Company, 2023).

Market Benchmarks

- Global telecom avg. churn: 20–35% annually (GSMA, 2024)
- Company A observed churn rate: 49.6% — well above benchmark
- Each churned customer = lost CLV of ~\$1,106 (avg. in dataset)
- Early intervention can reduce churn by 10–25% (McKinsey, 2023)

49.6%

Company A
Churn Rate

20–35%

Industry
Average Churn

5–7×

Cost: Acquire
vs Retain

\$1,106

Avg. Customer
Lifetime Value

Dataset Overview & Exploratory Analysis

100,000

Total Customers

100

Features

2 Files

Client + Record

49.6%

Churn Rate

Key EDA Findings

Equipment Age (eqpdays)

Customers with devices >24 months old churn at rates 5–8% higher than average. Outdated hardware drives dissatisfaction.

Declining Usage (change_mou)

Customers with usage drop >200 MoU/month show churn rates exceeding 60%. Disengagement is an early warning signal.

Customer Care Calls

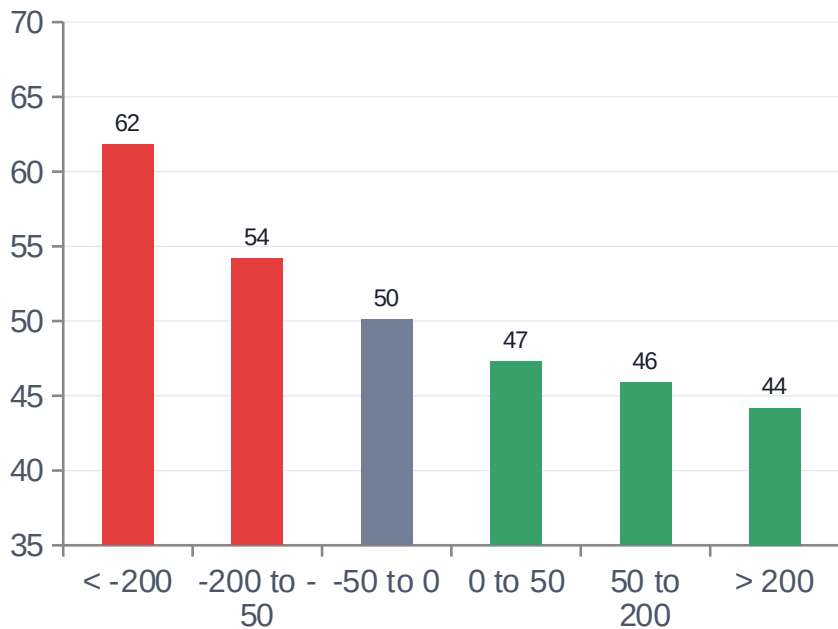
Churners average 1.6 care calls/month vs 2.0 for retained customers — indicating they stop seeking help before leaving.

Low Completion Rate

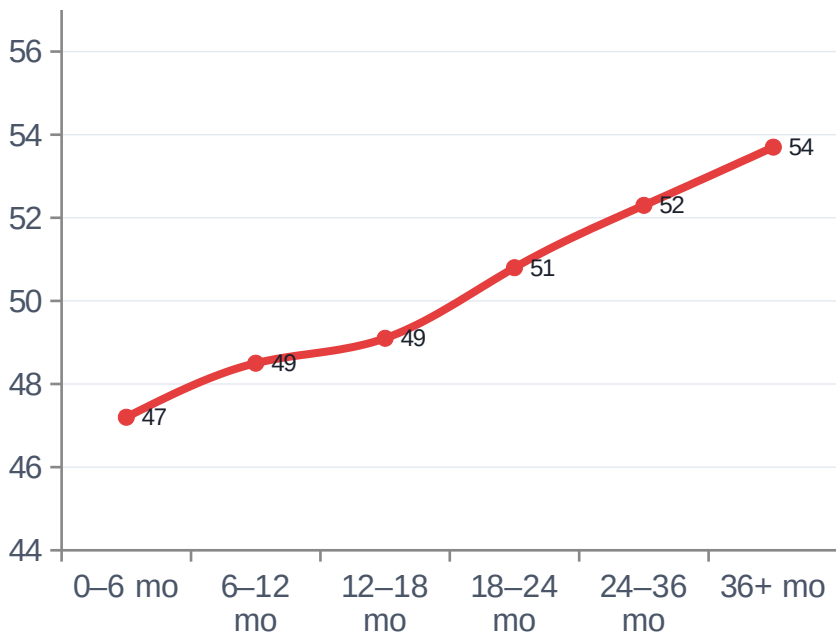
Churners show lower call completion ratios, correlating with service quality dissatisfaction.

Churn Patterns in the Data

Churn Rate by Monthly Usage Change



Churn Rate by Equipment Age



Machine Learning Model Comparison

Logistic Regression

57.65%

Accuracy

0.6061

AUC-ROC

Metric: AUC-ROC (area under ROC curve)

Linear baseline. Fast and interpretable but limited in capturing non-linear patterns in this dataset.

Random Forest

62.16%

Accuracy

0.6698

AUC-ROC

Metric: AUC-ROC (area under ROC curve)

Ensemble of 100 decision trees. Good performance and robust feature importance ranking.

✓ SELECTED

Gradient Boosting

62.12%

Accuracy

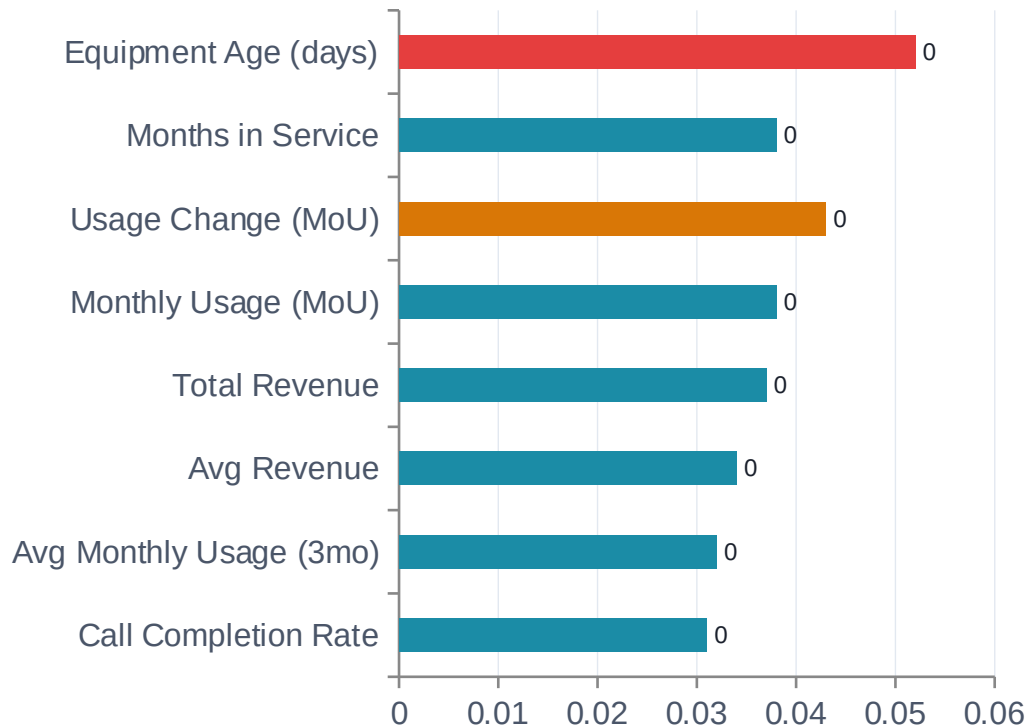
0.6768

AUC-ROC

Metric: AUC-ROC (area under ROC curve)

Sequential boosting approach. Highest AUC-ROC. Selected as the production model.

Key Churn Drivers (Feature Importance)



Model: Random Forest (100 estimators) | Metric: Gini Importance

Business Interpretation

Equipment Age

Top predictor. Older devices directly link to customer dissatisfaction and switching intent.

Usage Change

Sharp declines in monthly usage (MoU) signal disengagement 30–60 days before churn.

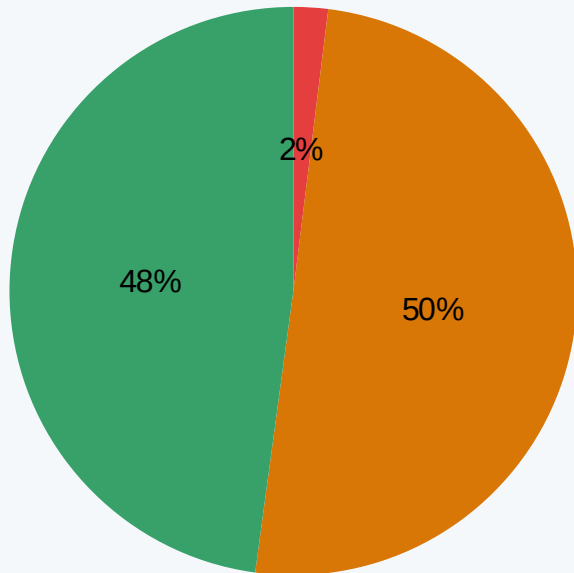
Service Months

Newer customers and very long-tenured customers both show churn risk — for different reasons.

Revenue Metrics

Lower-revenue customers are more price-sensitive; sudden revenue drops also signal risk.

Customer Risk Segmentation



■ High Risk ($\geq 75\%$) ■ Medium Risk (50–75%)
■ Low Risk ($< 50\%$)

Intervention Strategy by Tier

High Risk (2.0% • ~1,968 customers)

Immediate proactive outreach + device upgrade subsidy offer. Personal call from retention team.

Medium Risk (50.2% • ~50,182 customers)

Personalized loyalty reward + plan optimization. Targeted SMS/email campaign.

Low Risk (47.8% • ~47,850 customers)

Engagement programs + NPS surveys. Periodic value reminders.

Our Solution:

Predictive Churn Prevention Platform



Detect

Real-Time Scoring

Deploy GB model as API. Score all customers weekly. Flag high-risk accounts before they churn.



Target

Segmented Outreach

Route each risk tier to the right retention channel: personal call, SMS campaign, or loyalty reward.



Measure

Continuous Learning

Track retention rates post-intervention. Re-train model monthly with new behavioral data.

Financial Impact & ROI Estimate

Assumptions

- Avg. monthly revenue/customer: \$58.72
- Avg. customer tenure: 18.8 months
- Customer Lifetime Value (CLV): \$1,106
- High-risk customers flagged: ~1,968
- Assumed intervention success rate: 15%
- Intervention cost per customer: \$50

Estimated Outcomes (High-Risk Tier)

295

Customers Retained

\$0.21M

Annual Revenue Saved

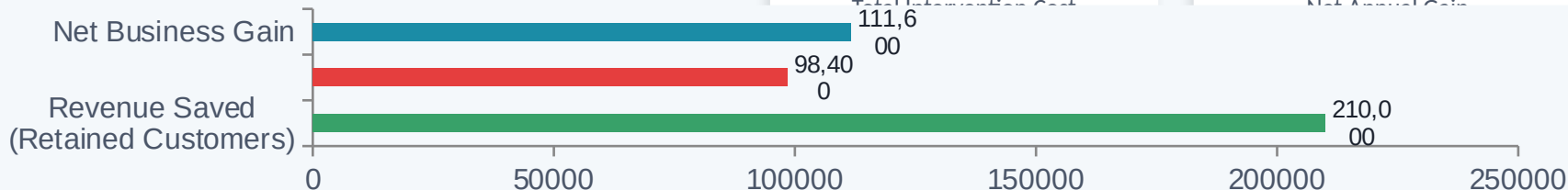
\$0.10M

Total Intervention Cost

\$0.11M

Net Annual Gain

Annual Revenue: Scenario Comparison



Implementation Roadmap

01

Month 1-2

Foundation

- Deploy GB model as REST API
- Build real-time risk scoring pipeline
- Integrate with CRM system

02

Month 3-4

Intervention

- Launch high-risk outreach campaign
- Introduce device upgrade subsidy for eqpdays > 24mo
- Activate personalized loyalty rewards

03

Month 5-6

Evaluation

- A/B test campaign effectiveness
- Track retention lift vs. control group
- Measure ROI vs. targets

04

Month 7+

Scale & Refine

- Expand to medium-risk tier
- Automate monthly model re-training
- Explore deep-learning uplift models

Summary & Next Steps

Business Problem	49.6% churn rate — 2× industry benchmark
Key Drivers	Equipment age (eqpdays), declining usage (change_mou), low engagement
Best ML Model	Gradient Boosting — AUC-ROC: 0.6768 Accuracy: 62.12%
Evaluation Metric	AUC-ROC (measures ranking quality across all thresholds)
Proposal	3-tier AI-powered churn prevention platform
Est. Net Gain	\$0.11M/year on high-risk tier alone (conservative estimate)
Next Step	PoC pilot: deploy scoring API + retention campaign on high-risk segment

References

- [1] GSMA Intelligence (2024). The Mobile Economy 2024. <https://www.gsma.com/mobileeconomy/>
- [2] Bain & Company (2023). Putting the Service Profit Chain to Work. Harvard Business Review.
- [3] McKinsey & Company (2023). Reducing Churn in Telecom: The Analytics Playbook.
- [4] Scikit-learn Developers (2024). Scikit-learn: Machine Learning in Python. <https://scikit-learn.org>
- [5] Company A Dataset — GCI World 2026 April Final Assignment (provided materials).
- [6] Matsuo-Iwasawa Lab UTokyo. GCI World 2026 April Course Materials.

Note: All AI-assisted conversations used in this assignment should be cited with their URL in the Omnicampus submission form.